

# F37 — Substrate-Schur generator P (Hardy projection) FORWARD framework v0.1: one operator, two sectors (muon edge $81/8 + \Lambda$ vacuum factor $\approx 2$ )

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## F37 Substrate-Schur Generator P — FORWARD Framework

### 0. Goal and scope

Phase 5 dual-track first content. The morning (F32/F35) produced a candidate cross-link: the muon mass-ratio's "edge" term and the cosmological  $\Lambda$  over-prediction are the *same bulk $\oplus$ Shilov partition* of  $D_{IV}^5$ . Cal's discipline (Cal #254: share an operator, not an integer) is the bar this must clear. This note proposes the shared operator explicitly — the Hardy/Bergman projection P — and takes the first derivation step (the vacuum sector), which is the cleanest. The muon sector is set up but not closed. Tier: FRAMEWORK candidate; the operator is concrete, the shared geometric ratio  $\rho$  is identified, and computing  $\rho$  is the open core.

### 1. The candidate generator: P

Let  $L^2(D_{IV}^5, d\mu)$  be the substrate Hilbert space and P the orthogonal projection onto the Hardy/Bergman space  $H^2(D_{IV}^5)$  (holomorphic  $L^2$ , the Tier 0 commitment space).  $(1-P)$  projects onto the complement. P is the proposed substrate-Schur generator in Cal #36's sense: one operator whose matrix elements in distinct physical sectors generate distinct observables. The shared quantity across sectors is not an integer — it is a geometric ratio  $\rho$  intrinsic to P (Sec. 3). That is what makes this the right *shape* of candidate.

## 2. First derivation step — the vacuum sector

### 2.1 The vacuum splits additively (the one thing proved here)

The substrate vacuum energy is the regularized trace of  $H_B$  = Casimir of  $K = SO(5) \times SO(2)$  over  $L^2$ . Split by  $P$ :

$$E_{\text{vac}} = \text{Tr}_{L^2}(H_B) = \text{Tr}(PH_BP) + \text{Tr}((1-P)H_B(1-P)) + 2 \text{Re} \text{Tr}(PH_B(1-P)).$$

Claim: the cross term vanishes, because  $[H_B, P] = 0$ . Proof:  $H_B$  is the Casimir of  $K$ , hence central in the universal enveloping algebra of  $\mathfrak{k}$  — it is  $K$ -invariant.  $H^2(D_{IV}^5)$  is a  $K$ -invariant subspace (the holomorphic  $K$ -types), so its complement is  $K$ -invariant too. A  $K$ -invariant operator commutes with the projection onto any  $K$ -invariant subspace. Therefore  $[H_B, P] = 0$ , the cross term is zero, and

$$E_{\text{vac}} = E^{\text{bulk}} + E^{\text{bdy}}, \quad E^{\text{bulk}} = \text{Tr}(PH_BP), \quad E^{\text{bdy}} = \text{Tr}((1-P)H_B(1-P)).$$

This is a genuine derivation, not a scaffold: the vacuum energy is *exactly* additive over the bulk/boundary partition, with no cross-contamination, because the time-generator commutes with the partition. That is the operator-level content the cross-link needs.

### 2.2 The factor is $1 + \rho$

Per Casey's 2-region insight, the observer is bulk-localized and measures only  $E^{\text{bulk}}$  (the Shilov-boundary vacuum is subtracted at the observer's location); the substrate carries both. Hence

$$\frac{\Lambda_{\text{substrate}}}{\Lambda_{\text{observed}}} = \frac{E^{\text{bulk}} + E^{\text{bdy}}}{E^{\text{bulk}}} = 1 + \rho, \quad \rho \equiv \frac{E^{\text{bdy}}}{E^{\text{bulk}}}.$$

The observed factor is  $4.85/2.4 = 2.0208$ , so the framework predicts  $\rho = 1.0208$  — the boundary and bulk vacuum energies are nearly equal. This is a forward, falsifiable prediction:  $\rho$  is a fixed geometric invariant of  $D_{IV}^5$  (the ratio of the Hardy/Shilov normalization to the Bergman/bulk normalization of the same holomorphic spectral data), computable from  $c_{FK}$  and the Shilov measure. Compute  $\rho$ ; if  $\rho \neq 1.02$  (within tolerance), the bulk $\oplus$ Shilov vacuum reading fails. The honest content of “factor  $\approx 2$ ” is  $\rho \approx 1$ , with the residual 0.02 being the genuine value of the normalization ratio — not a fitted  $81/40$  (F35 declined that).

### 2.3 Subtlety to settle in the rigorous pass (flagged)

Whether  $E^{\text{bdy}}$  and  $E^{\text{bulk}}$  are contributions from *orthogonal* regions or two *normalizations* of the same Hardy data (Bergman bulk  $\leftrightarrow$  Shilov boundary are isomorphic realizations of  $H^2$ ) changes the interpretation of  $\rho$  but not the structural form

factor =  $1 + \rho$ . The rigorous computation must settle which; I flag it rather than assert it.

### 3. The shared ratio $\rho$ enters the muon sector (the unification, set up not closed)

The muon edge-term claim (F32 Gate A1): the  $(1-P)$  boundary matrix element of the gen-2 mass operator equals  $N_c^4/2^{\{N_c\}} = 81/8$  relative to the bulk normalization. The key structural point: the  $(1-P)$ -vs- $P$  matrix element of *any* operator on a K-type is governed by the same Bergman-to-Hardy normalization ratio  $\rho$  that controls the vacuum sector, modulated by the K-type's own weight. Schematically,

$$\langle V_{(3/2,1/2)} | (1-P) M (1-P) | V_{(3/2,1/2)} \rangle = \rho \cdot \kappa(V_{(3/2,1/2)}),$$

where  $\kappa$  is the K-type form-factor (Pochhammer/Casimir structure of Ch 5/Ch 7). If this holds with the same  $\rho$  as Sec. 2.2, then  $P$  is genuinely one operator generating both observables — the vacuum factor  $(1+\rho)$  and the muon edge  $(\rho \cdot \kappa)$  — and the cross-link is operator-level, exactly the Cal #254 contrast class. The shared object is  $\rho$  (geometric), never the integer 81.

What is NOT done: I have not computed  $\rho$ , nor  $\kappa(V_{(3/2,1/2)})$ , nor shown  $\rho \cdot \kappa = 81/8$ . Those are the open core. Today establishes only: (a) the vacuum splits additively because  $[H_B, P]=0$ ; (b) the factor has the forced form  $1+\rho$ ; (c) the *same*  $\rho$  is the natural object in the muon sector. That is enough to make  $P$  a FRAMEWORK candidate generator and to scope the remaining work sharply.

### 4. Honest status and falsifiers

- Established (Tier: derived):  $[H_B, P] = 0$  [?] additive bulk/boundary vacuum split; factor =  $1 + \rho$  exactly.
- Forward prediction (falsifiable):  $\rho = 1.02$  (geometric Bergman/Hardy normalization ratio). Compute it;  $\rho \neq 1.02$  refutes.
- Candidate (FRAMEWORK):  $P$  is a substrate-Schur generator; the same  $\rho$  generates the muon edge-term  $\rho \cdot \kappa = 81/8$ . Falsifier: if the muon sector requires a *different* ratio  $\rho_2 \neq \rho$ , the operator is not shared and the cross-link drops to coincidence.
- Declined (F35): the 81/40 numerical match for 2.02 — superseded by the mechanism  $\rho \approx 1$ .

### 5. Multi-week core (Cal #189), now sharply scoped

1. Compute  $\rho = E^{\text{bdy}}/E^{\text{bulk}} = \text{Bergman/Hardy normalization ratio of } D_{IV}^5$  from  $c_{FK} + \text{Shilov measure}$ . (Closes the vacuum prediction.)
2. Compute  $\kappa(V_{(3/2,1/2)})$  and test  $\rho \cdot \kappa = 81/8$ . (Closes the muon edge / Gate A1.)

3. Settle the orthogonal-vs-normalization subtlety (Sec. 2.3).
4. Cal cold-read + Keeper K-audit (K229d against A1).

## 6. Closure

F37 v0.1: P (the Hardy/Bergman projection) proposed as a substrate-Schur generator. The vacuum sector is taken one real step forward —  $[H_B, P]=0$  forces an additive bulk/boundary vacuum split, giving the factor the exact form  $1+\rho$  with the forward prediction  $\rho \approx 1.02$ . The muon sector is set up under the *same*  $\rho$ , making the cross-link operator-level (Cal #254), with the open core (compute  $\rho$ ; test  $\rho \cdot \kappa = 81/8$ ) sharply scoped. The shared object is the geometric ratio  $\rho$ , never the integer 81 — the F35 discipline carried into the FORWARD framework.

Tier: F37 substrate-Schur generator P FORWARD framework v0.1;  $[H_B, P]=0$  additive split DERIVED;  $\rho \approx 1.02$  forward prediction; P-as-generator FRAMEWORK candidate; multi-week core scoped per Cal #189.

— Lyra, Sat 2026-06-06 12:15 EDT. F37 v0.1: Hardy/Bergman projection P proposed as substrate-Schur generator unifying muon edge  $(81/8) + \Lambda$  vacuum factor ( $\approx 2$ ); first derivation step —  $[H_B, P]=0$  ( $H_B$  K-invariant Casimir commutes with projection onto K-invariant  $H^2$ )  $\Rightarrow$  vacuum splits additively  $E^{\text{bulk}+E^{\text{bdy}}}$ , cross term zero  $\Rightarrow$  factor  $= 1+\rho$  exactly, forward prediction  $\rho = 1.0208$  (Bergman/Hardy normalization ratio, geometric invariant — compute to falsify); same  $\rho$  enters muon sector as  $\rho \cdot \kappa(V_-(3/2, 1/2)) = 81/8$  (set up, not closed); shared object is  $\rho$  not integer 81 (Cal #254 contrast class, F35 discipline carried forward); open core = compute  $\rho$  + test  $\rho \cdot \kappa = 81/8$ , multi-week per Cal #189.